



An RF-CNN pipeline for predicting PM_{2.5} concentration in Sri Lanka

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ABSTRACT

Air pollution is a considerable global public health threat, requiring efficient monitoring and forecasting to guide decision-making. This study introduces a cascaded model of enhanced Random Forest with Convolutional Neural Network (RF-CNN) that predicts spatiotemporal fluctuations in PM_{2.5} concentrations throughout Sri Lanka. The K-Nearest Neighbors method is employed to impute missing data, and the model utilizes data from 24 low-cost PM_{2.5} sensors that are distributed throughout the country. The Convolutional Neural Network (CNN) derives spatial features from four-band PlanetScope satellite images (3m/pixel resolution, 1km² spatial coverage), while the Random Forest (RF) component models the relationship between PM_{2.5} levels and four meteorological parameters. These features, combined with meteorological, spatial, and temporal inputs, produce the final forecasting results. The dataset comprises 1934 satellite images that were collected between December 2022 and February 2024, with an average PM_{2.5} concentration of approximately 15 µg/m³. The RF-CNN model exhibited robust performance metrics across a variety of climate zones, including a normalized root mean square error of approximately 32.4 %, a mean absolute percentage error of approximately 25.7 %, a normalized mean absolute error of approximately 22.8 %, a Spearman r of 0.871, and a Pearson r of 0.873. Two metrics: Input Data Quality Score (IDQS) and Testing Data Quality Score (TDQS) were implemented to evaluate the effects of imputation. Performance was minimally impacted by imputation within acceptable ranges, while exceeding limits resulted in increased uncertainty. This research emphasizes the efficacy of the RF-CNN approach, which integrates satellite imagery and low-cost sensor data, as a scalable solution for predicting spatiotemporal PM_{2.5} variations. It provides valuable insights for regions that lack extensive monitoring.

1. Introduction

The inhalation of fine particulate matter in the atmosphere has been linked to adverse health effects, particularly in the context of cardiovascular and respiratory diseases (Pope and Dockery, 2006). A lack of awareness concerning the deleterious effects of fine particulate matter and the inadequacy of emission controls in densely populated areas worldwide has led to exceedingly high levels. The elevated levels of PM not only give rise to issues in the areas where they originate, but also have a cascading effect on countries and regions situated at a distance. In 2019, polluted air caused an estimated 4.2 million deaths globally (Ambient (outdoor) air pollution). The Global Burden of Disease 2021

report identified ambient particulate matter pollution as a primary risk factor that significantly contributes to the global burden of disease (Institute for Health Metrics and Evaluation, 2024). According to the 2023 World Air Quality Report, South Asia has the most polluted cities in the world (World Health Organization, 2023). Air pollution is the third-highest risk factor for premature deaths in South Asia (World Bank, 2023). Although the South Asian region shows high levels of PM_{2.5}, Sri Lanka has comparatively low levels of PM_{2.5} (Jabbar et al., 2022).

Even though Sri Lanka has low levels of ambient PM_{2.5} compared to the region, the levels are high compared with World Health Organization (WHO) standards (Rathnayake et al., 2023). Also, the country

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experiences a seasonal effect of air pollution, with elevated levels occurring during the dry season, particularly those observed in the Bay of Bengal, which also exert an influence on air quality in the island (Abeyratne and Ilerperuma, 2006; Senarathna et al., 2024). It is observed that the country is subjected to transboundary air pollution, yet the extent and impact of this phenomenon remain poorly understood. To prepare for future challenges, it is essential to conduct thorough research and analysis on the complex relationships between meteorological conditions, pollution levels, and other contributing factors, in order to develop effective strategies for mitigating potential adverse impacts of air pollution ().

Forecasting $PM_{2.5}$ concentration is an important aspect of preventing air pollution and taking action to improve air quality. Many forecasting models are developed to predict $PM_{2.5}$ concentration based on statistical and artificial intelligence models (Liu et al., 2021; Donnelly et al., 2015). Statistical models show limitations in learning complex, random, and nonlinear time series data (Rybarczyk and Zalakeviciute, 2018). Artificial Intelligence based models are much more capable of learning complex nonlinear time series data. Support Vector Machines (SVM) (Castelli et al., 2020), Artificial Neural Networks (ANN) (Maleki et al., 2019), and Deep Learning models such as Recurrent Neural Networks (RNN), Long Short-Term Memory Networks (LSTM) (Li et al., 2017), and Gated Recurrent Unit networks (GRU) (Zhou et al., 2019) are the popular proposed methods for predicting $PM_{2.5}$ concentrations. Various hybrid and powerful models have been proposed to enhance the forecasting capabilities of the prediction model. Duan J. et al. proposed an ARIMA-CNN-LSTM combination model to predict air quality and used the Dung Beetle optimizer algorithm to find the optimal hyperparameters of the CNN-LSTM part (Duan et al., 2023). Zhang Z. et al. introduced a new lightweight deep learning model using a sparse attention-based transformer network to capture long-term dependencies and complex relationships from $PM_{2.5}$ time series (Zhang and Zhang, 2023). Chen H. et al. proposed a prediction method based on an integrated dual-LSTM model that uses an eXtreme Gradient Boosting (XGBoosting) tree to integrate two LSTM models (Chen et al., 2021). However, these high-performing models need high computational power and take longer to learn the dynamics. Prediction interval is also a vital concern in the $PM_{2.5}$ forecasting models since the predictor needs short-term as well as long-term prediction to prevent adverse $PM_{2.5}$ conditions, adjust the environmental policies of countries, and take global actions against air pollution.

Despite the limited information available on the spatial distribution of $PM_{2.5}$ due to the scarcity of regulatory Air Quality Monitoring (AQM) stations, low-cost AQM systems can effectively address this issue by allowing for the exploitation of the significant spatial variations in $PM_{2.5}$ concentrations. Machine learning and statistical models are usually used in combination with satellite data to estimate PM concentrations at higher spatial resolution, especially below 1 km (Yao et al., 2019; Zheng et al., 2020; Zheng et al., 2021; Jiang et al., 2022). Zheng T. et al. proposed a Convolutional Neural Network – Random Forest (CNN-RF) cascaded model where the CNN part extracts feature from the satellite imagery and RF regressor non-parametrically models the association of $PM_{2.5}$ concentrations and the extracted features (Zheng et al., 2020). Continuing their previous studies, Zheng T. et al. discussed the disadvantages of the CNN-RF approach and introduced an RF-CNN cascaded model that utilizes RF to predict $PM_{2.5}$ levels generally with low prediction errors and CNN to predict the residual spatial variabilities of $PM_{2.5}$ concentrations utilizing satellite imagery. The piggybacked CNN model on low RF prediction errors successfully predicted the residual for the Delhi and Beijing Datasets (Zheng et al., 2021). Jiang Z. et al. developed a new framework called Spatiotemporal Contrastive Learning (SCL) for precisely assessing the spatial variations of ground-level $PM_{2.5}$ concentrations using satellite imagery (Jiang et al., 2022). Despite the precise timing of satellite images, they are more advantageous for predicting 24-hour aggregated targets because of the reduced measurement noise (Zheng et al., 2020).

The current study aims to assess the applicability of the RF-CNN model in predicting $PM_{2.5}$ concentrations in countries such as Sri Lanka, where the average $PM_{2.5}$ concentrations are comparatively consistently low compared to the South Asian region and how to tackle the lack of ground-truth data in the sensor network to predict the $PM_{2.5}$ concentrations. Hence, comparative studies and ablation studies were not conducted. We made certain modifications to its original approach. The RF component functions as an error model, effectively manipulating the remaining $PM_{2.5}$ concentrations so that the CNN component can make predictions with minimal errors. CNN utilizes 15 spatiotemporal features and meteorological features with satellite images to forecast the final results. The main contributions of this study are; (1) Ground-level $PM_{2.5}$ data collection and filling the data gaps with the KNN-imputation method, (2) Satellite image collection and preprocessing, (3) Training the RF model with meteorological features and fine-tuning it through cross-validation, and (4) Training the RF-CNN cascaded model and evaluating the results using appropriate error metrics. The proposed forecasting model effectively predicts daily average $PM_{2.5}$ concentrations in different cities in Sri Lanka. This study also discusses the application and constraints of the KNN imputation method for enhancing the model outcomes. In addition, we provide a brief discussion on the sensitivity of satellite images in predicting ground-level $PM_{2.5}$ concentrations.

2. Materials and methods

2.1. Ground-level $PM_{2.5}$ data collection

Hourly ground-level $PM_{2.5}$ concentration readings were gathered from selected 24 AQM sites scattered across Sri Lanka. Each AQM station has a TSI BlueSky 8143 air quality monitors mounted at least 1.5 m above ground level. Fig. 1 shows the established BlueSky sensors at four different AQM stations. The BlueSky 8143 sensors from TSI are low-cost monitors that measure $PM_{2.5}$ concentration, temperature, and relative humidity in real-time. It is suitable for outdoor usage and include Wi-Fi and cellular modem connection capabilities. The TSI Link enables complete data access via its cloud platform.

This study implemented a season wise calibration strategy to increase the accuracy of the low-cost $PM_{2.5}$ sensor accuracy. Initially, sensor harmonization was performed to correct systematic biases among devices. Linear Regression (LR), a simple but very robust model is chosen as the temporal calibration method to develop the calibration model using co-located measurements from reference-grade BAM-1020 monitors stationed at the American Club in Colombo and Torrington Park in Kandy, Sri Lanka. Two distinct calibration models were developed for the wet season (May-September) and the dry season (October-April) in Sri Lanka. In calibrating daily average $PM_{2.5}$ measurements, a complex approach was unnecessary as linear regression yields sufficient results. Fig. 2 displays the distribution of all 24 AQM sites across Sri Lanka. These AQM stations were strategically positioned to collect data on the levels of $PM_{2.5}$ that the majority of the general population is exposed to. Each data entry is spaced 15 min apart and contains the date-time, $PM_{2.5}$ concentration ($\mu g/m^3$), relative humidity, and temperature in Celsius. The $PM_{2.5}$ data series included in this study spans from the 01st of November 2022 to the 29th of February 2024.

The hourly average $PM_{2.5}$ concentration is determined by computing the mean of four data entries within each hour. Due to a potential sensor device failure, power failure within the sensor location, or Wi-Fi issues preventing data transfer to the cloud, every station had missing data. Hence, to consider the hourly average sufficiently precise, at least three data entries within a one-hour time frame must be recorded. Nevertheless, employing a suitable imputation method on hourly $PM_{2.5}$ time series can effectively address the issue of missing data. Moreover, the impact of this imputation process on the daily $PM_{2.5}$ time series would be insignificant, as it would primarily arise from taking the 24-hour mean of the hourly $PM_{2.5}$ time series. Acquiring the complete daily

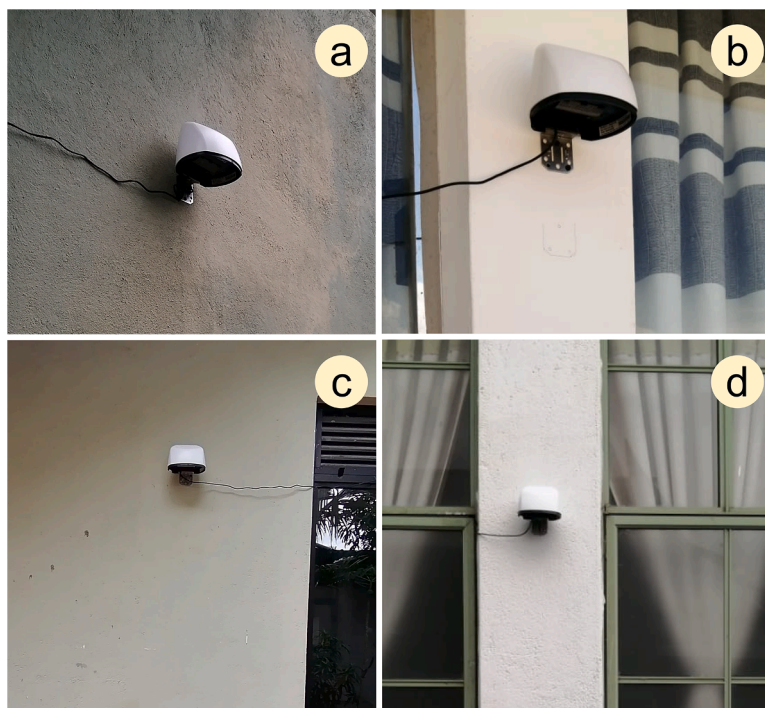


Fig. 1. TSI BlueSky Sensors established at a) Central Environmental Authority (CEA) office - Kanthale; b) District Secretariat (DS) Office - Mullaitivu; c) CEA office - Ampara; and d) DS office - Nuwaraeliya.

PM_{2.5} time series is crucial for effectively training the prediction model.

The available hourly data was utilized for addressing the missing data by using K-Nearest Neighbour (KNN) imputation. Complex imputation methods were not utilized due to the requirement of substantial clean data to implement and overfitting risk. The utilization of a substantial quantity of k-neighbors may seem advantageous in nearest-neighbor imputation due to the perceived improvement in accuracy. However, this improvement conceals a significant alteration in the actual distribution of data (Beretta and Santaniello, 2016). Therefore, we employed a moderate value for the k parameter ($k = 5$) in the KNN imputation method. Hour of the day, day of the week, day of the month, and month of the year were used as features of the KNN imputation model, and the imputation is conducted on a station-by-station basis. The imputation process has effectively filled the data gaps while preserving the information of non-meaning data. As shown in Fig. 3, the distribution of the total imputed hourly PM_{2.5} values closely resembles the distribution of PM_{2.5} values in the raw hourly PM_{2.5} data. After imputation, the dataset exhibits a comparable hourly PM_{2.5} distribution. Subsequently, a 24-hour aggregated dataset of PM_{2.5} values was generated for each AQM station.

To fully establish a data point, it is necessary to have the corresponding satellite images, meteorological data, and spatiotemporal features corresponding to the ground-truth daily average PM_{2.5} concentrations. Hence the number of data points is reduced due to the limited availability of satellite images, meteorological data, and other spatiotemporal features. Fig. 2 illustrates the allocation of the 24 AQM sites into a 17:7 ratio for the training set and testing set while maintaining similar summary statistics in both sets. Training set and test set cover almost every climate zone in Sri Lanka. Table 1 includes the geographic coordinates of the AQM sites, as well as the average, standard deviation, and range of the ground-truth PM_{2.5} concentrations. It also indicates whether each site is part of the training set or the testing set. Table 2 displays the summary statistics and the number of data points for both the train and test sets.

2.2. Satellite image collection

The PlanetScope images served as the primary data source for collecting the satellite images. PlanetScope satellite imagery is captured in the form of a continuous strip of individual frame images referred to as "scenes". In this study, we have utilized the 'ortho' geometry type and the 'ortho_visual' asset type for the images which have 4 bands (blue, green, red, and near-infrared) and 3 m/pixel resolution. The PlanetScope Ortho Scene product undergoes orthorectification, processing to eliminate distortions resulting from terrain and is better for applications demanding imagery with precise geo-location and cartographic projection. Since we were only interested in estimating PM_{2.5} at sub-km resolutions from satellite images of spatial coverage no bigger than 1 km x 1 km, we established our areas of interest (AOI) as the 24 1 × 1 km square grid cells, with each of the 24 AQM stations positioned at the geographic midpoint of the square grid. Then, a search and download operation was conducted for any images that overlapped with our AOI over the specified period range. Upon cropping the images (224 × 224 pixels) to obtain our AOI, we manually removed images with high cloud coverage and higher percentages of darker pixels. It was straightforward since our image collection is relatively small. Then we perform the filtration process once more using a cloud coverage filter to exclude any images with cloud coverage exceeding 10 %. Manual filtration was necessary for removing the images that are affected by cloud shadows and contains limited information. This ensured that the CNN concentrated on significant patterns significant to the Area of Interest (AOI), enhancing feature extraction and mitigating the risk of misleading inputs. As a result, we have successfully generated a collection of images that contain a minimal number of cloudy images or images with limited information such as images with a high proportion of dark pixels caused by cloud shadows. Fig. 4 displays examples of cloudy images and images that have limited information on them. Each satellite image was aligned with its respective station's ground-truth daily average PM_{2.5} concentration, meteorological data, and spatiotemporal data. Table 1 shows the total number of satellite images available for each site. Note that, >1 image is selected for a day if the available images are captured at least 10

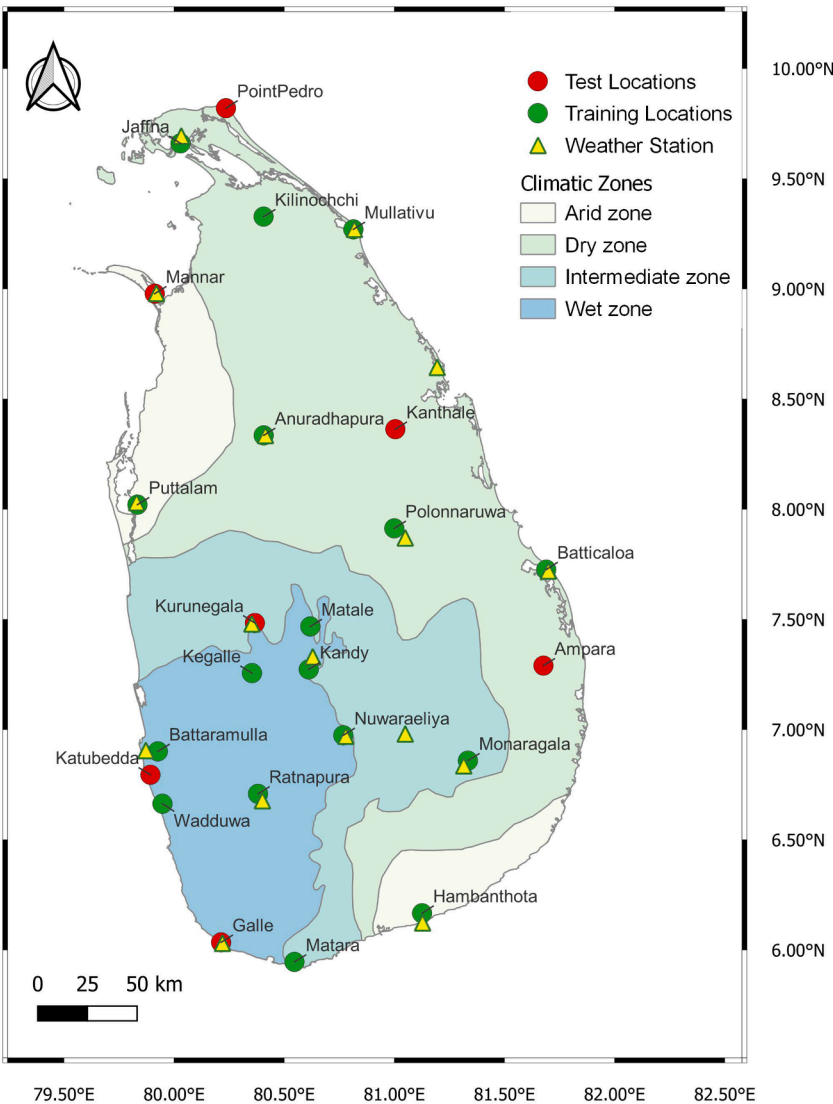


Fig. 2. Locations of selected 24 AQM stations and weather stations in Sri Lanka where red dots indicate training stations, green dots indicate testing stations utilized for the RF-CNN model in this study, and yellow triangles indicate weather stations used to collect meteorological data.

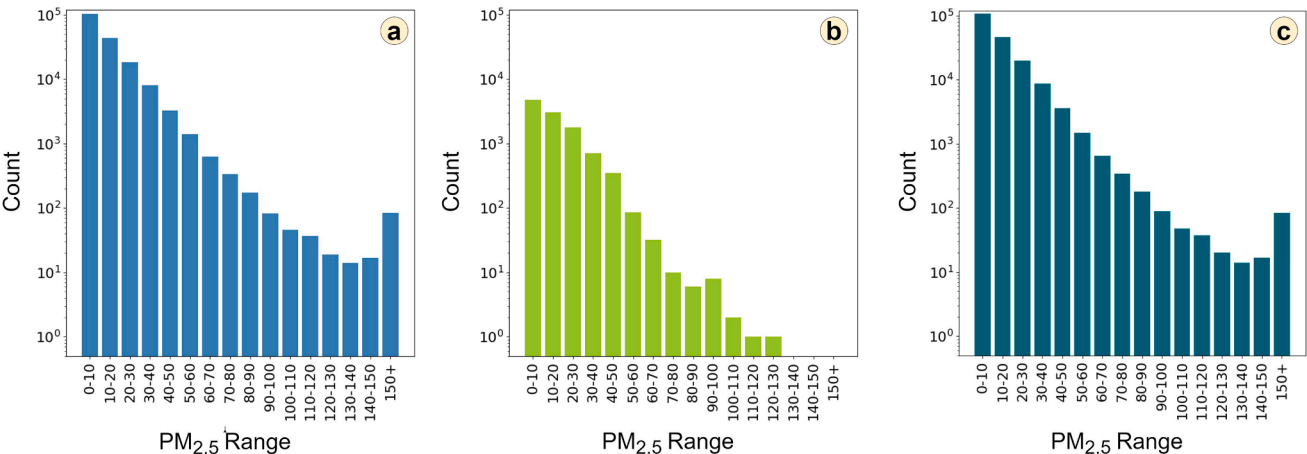


Fig. 3. Histograms of a) Total hourly raw $PM_{2.5}$ concentrations ($\mu g/m^3$); b) Total KNN imputed $PM_{2.5}$ concentrations ($\mu g/m^3$); and c) Total $PM_{2.5}$ concentrations after KNN imputation ($\mu g/m^3$). The y-axis of each histogram plot is represented on a logarithmic scale.

Table 1

The site name, coordinates, data availability period, the number of data points, summary statistics, and the category of the site for each of the 24 AQM stations across Sri Lanka.

Site	Lat	Lon	Data Availability Period	Data Points	Mean $\pm$ Std Dev ($\mu\text{g}/\text{m}^3$)	Range ($\mu\text{g}/\text{m}^3$)	Cat.
Kandy	7.2731	80.6117	2022/11/02 – 2024/02/29	164	19.49 $\pm$ 9.32	4.99 - 52.09	TRAIN
Galle	6.0339	80.2137	2023/03/29 – 2024/02/29	113	12.41 $\pm$ 7.88	2.68 - 38.88	TEST
Matale	7.4677	80.6188	2023/02/16 – 2024/02/29	111	21.69 $\pm$ 10.39	7.44 - 51.87	TRAIN
Nuwaraeliya	6.9737	80.7686	2023/03/02 – 2024/02/29	86	11.13 $\pm$ 6.25	1.43 - 28.88	TRAIN
Kegalle	7.2560	80.3540	2023/03/11 – 2024/02/29	103	16.06 $\pm$ 8.20	4.37 - 35.67	TRAIN
Rathnapura	6.7081	80.3813	2023/03/11 – 2024/02/29	113	17.41 $\pm$ 8.72	4.87 - 41.58	TRAIN
Matara	5.9451	80.5468	2023/03/23 – 2024/02/29	87	15.73 $\pm$ 9.26	4.79 - 39.96	TRAIN
Mannar	8.9783	79.9125	2023/04/26 – 2024/02/29	56	15.28 $\pm$ 8.61	3.33 - 31.55	TEST
Jaffna	9.6607	80.0294	2023/04/27 – 2024/02/29	47	15.51 $\pm$ 11.84	3.71 - 44.10	TRAIN
Point-Pedro	9.8202	80.2358	2023/04/27 – 2024/02/29	78	14.10 $\pm$ 9.74	3.27 - 38.19	TEST
Kilinochchi	9.3280	80.4066	2023/04/27 – 2023/10/12	31	5.22 $\pm$ 2.29	2.41 - 11.69	TRAIN
Mullaitivu	9.2702	80.8143	2023/04/28 – 2024/02/29	76	11.88 $\pm$ 8.54	1.92 - 41.67	TRAIN
Kanthale	8.3632	81.0050	2023/05/11 – 2024/02/29	68	12.20 $\pm$ 9.13	3.48 - 48.56	TEST
Polonnaruwa	7.9132	81.0001	2023/05/11 – 2024/02/29	84	9.76 $\pm$ 6.17	2.21 - 29.38	TRAIN
Batticaloa	7.7256	81.6900	2023/05/12 – 2024/02/29	78	10.43 $\pm$ 6.30	3.68 - 44.76	TRAIN
Ampara	7.2901	81.6774	2023/05/12 – 2024/02/29	81	11.99 $\pm$ 4.84	3.99 - 26.42	TEST
Monaragala	6.8587	81.3345	2023/05/13 – 2024/02/29	94	8.53 $\pm$ 4.74	1.28 - 27.63	TRAIN
Hambantota	6.1661	81.1263	2023/05/15 – 2024/02/29	67	9.66 $\pm$ 8.22	2.83 - 42.39	TRAIN
Anuradhapura	8.3351	80.4071	2023/02/12 – 2024/02/29	64	11.80 $\pm$ 6.88	2.55 - 33.07	TRAIN
Katubedda	6.7945	79.8928	2023/08/25 – 2024/02/29	44	26.88 $\pm$ 12.40	7.34 - 64.71	TEST
Wadduwa	6.6630	79.9473	2022/12/15 – 2024/02/03	90	21.30 $\pm$ 11.93	4.30 - 70.37	TRAIN
Kurunegala	7.4835	80.3671	2022/12/13 – 2024/02/24	67	15.64 $\pm$ 7.48	3.58 - 35.81	TEST
Puttalam	8.0207	79.8338	2022/12/19 – 2024/02/29	94	15.18 $\pm$ 10.23	2.54 - 54.92	TRAIN
Battaramulla	6.9008	79.9264	2023/02/07 – 2023/11/21	38	19.73 $\pm$ 11.34	5.66 - 41.39	TRAIN

Table 2

Overall summary statistics of the train set and test set including first-quartile (Q_1), median (Q_2), and third-quartile (Q_3).

Set	Data Availability Period	Data Points	Mean $\pm$ Std Dev ($\mu\text{g}/\text{m}^3$)	Range ($\mu\text{g}/\text{m}^3$)	Q_1	Q_2	Q_3
Train	2022/11/02 – 2024/02/29	1427	14.91 $\pm$ 9.81	1.28 – 70.37	7.64	11.94	19.94
Test	2022/12/13 – 2024/02/29	507	14.58 $\pm$ 9.41	2.68 – 64.71	7.40	11.48	19.62

min apart. This can be helpful to assess the level of sensitivity of the proposed model to the features extracted from the satellite images. Fig. 5 displays example satellite images for each of the 24 stations, showcasing the diverse landscapes of Sri Lanka.

2.3. Meteorological data

This study employed temperature (T) measured in Celsius, relative humidity (RH), wind speed (WS) expressed in kilometers per hour, and mean sea level pressure (MSLP) measured in hectopascal (hPa). Wind speed (WS) and mean sea level pressure (MSLP) data were collected from 17 weather stations spread across Sri Lanka and operated by the Department of Meteorology, Sri Lanka. (see Fig. 2). Since the majority of weather stations had complete data coverage within the study window, data gaps were filled using the data from the closest weather station.

For T and RH measurements, there were two options; 1) Measurements from the BlueSky device and 2) Prediction results of the Modern-Era Retrospective analysis for Research and Applications, Version 2 (MERRA-2) at a spatial resolution of 0.5° latitude by 0.625° longitude. Data gaps in the measurements from the BlueSky device were filled using the same KNN imputation method which was previously used to fill the gaps of ground-level $\text{PM}_{2.5}$ concentrations. A correlation analysis was conducted to determine the optimal choice between two options. Only the data from the Kandy site was utilized, while reference data was

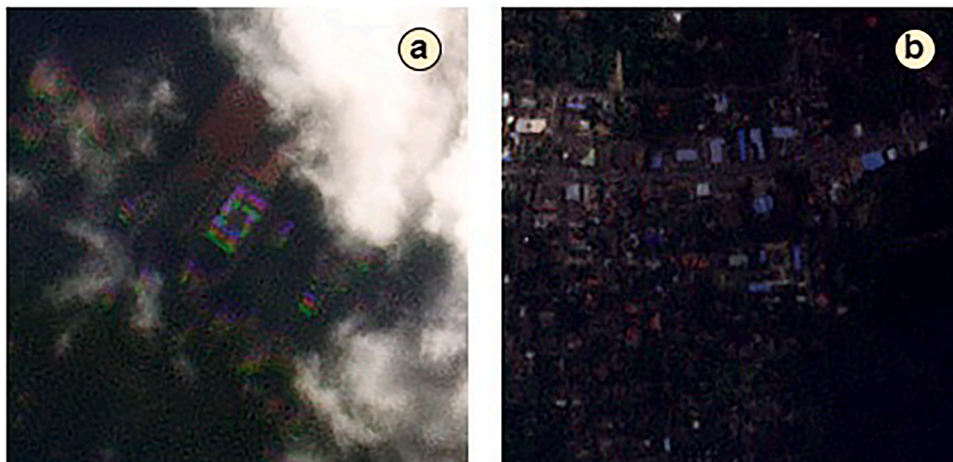


Fig. 4. Examples of filtered-out a) cloudy image and b) image that has low information.

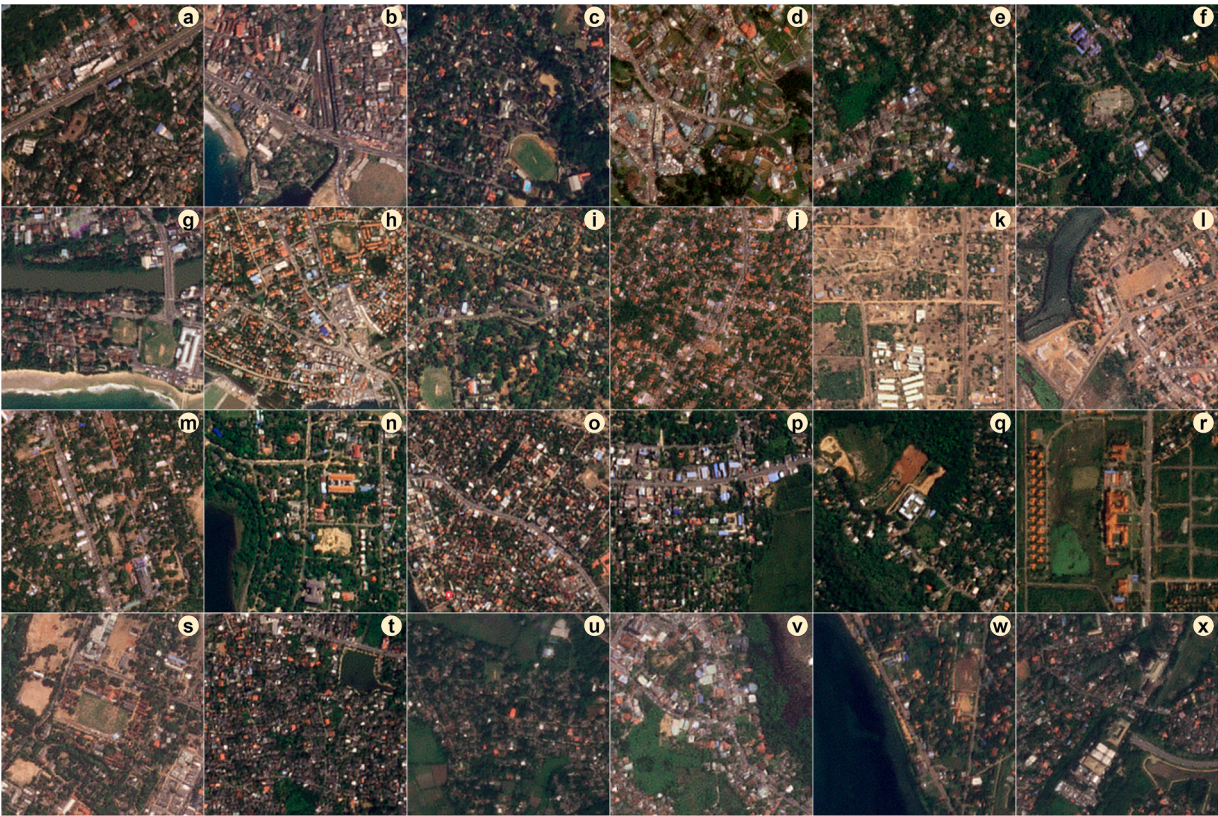


Fig. 5. Example satellite images for each 24 AQM stations; a) Kandy, b) Galle, c) Matale, d) Nuwaraeliya, e) Kegalle, f) Ratnapura, g) Matara, h) Mannar, i) Jaffna, j) Point-Pedro, k) Kilinochchi, l) Mullaitivu, m) Kanthale, n) Polonnaruwa, o) Batticaloa, p) Ampara, q) Monaragala, r) Hambantota, s) Anuradhapura, t) Katubedda, u) Wadduwa, v) Kurunegala, w) Puttalam, and x) Battaramulla.

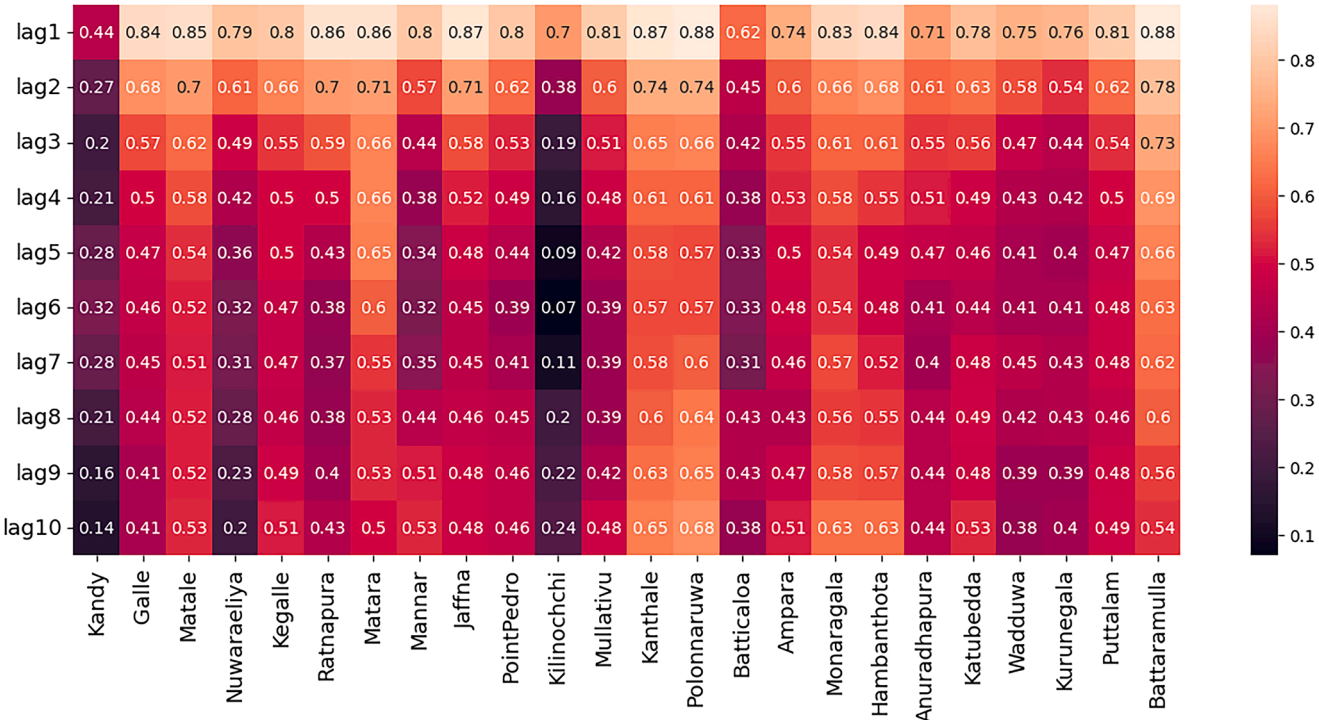


Fig. 6. Heat map of autocorrelation coefficients for the first 10 lags of each 24 daily average ground-level PM_{2.5} time series.

collected from the weather station located at the University of Peradeniya (latitude: 7.2558, longitude: 80.5979). Since measurements from the BlueSky device showed a higher Pearson correlation coefficient for both T and RH, it was chosen as the primary source for T and RH of the proposed model.

2.4. Spatiotemporal features

Ground-level daily average $PM_{2.5}$ concentrations exhibit a noticeable annual pattern. Therefore, the month of the year was chosen as an input for the proposed model. Having a complete set of daily average $PM_{2.5}$ time series for each AQM site enables us to utilize temporal characteristics in the proposed model. An auto-correlation analysis was performed to determine the optimal time lags to utilize in the proposed model as input features. Except for the Kilinochchi site, all other sites showed enough autocorrelation coefficients for the first 10-time lags, as depicted in Fig. 6. Note that the 5 % critical values of the autocorrelation under the null hypothesis of white noise are <0.16 for all the stations and all the lags. To keep the model simple, we have decided to incorporate 7-time lags; ground level $PM_{2.5}$ data from the past week into the proposed model as inputs. The reason for using time lags is that they also provide minor spatial information to the model by capturing the varying dynamics of $PM_{2.5}$ concentration among different AQM stations.

Population density is a reliable indicator for determining the level of

human activities occurring in a specific region. Human activities including cooking, operating vehicles, and incinerating waste directly or indirectly contribute to the rise in $PM_{2.5}$ levels in the nearby atmosphere. Also, the total road length and the total building area are good indicators for this purpose. Hence, OpenStreetMap (OSM), a free and open geographic database was used to collect spatial variations of population, total road length, and total building area. Population density, total road length, and total building area were calculated for each AQM site within a 1 km radius from its location.

2.5. RF-CNN pipeline

This study follows an extension of the methodology described by Zheng et al. (2021), with modifications made to the RF and CNN components. RF-CNN pipeline, as shown in Fig. 7, was jointly trained on satellite images, meteorological conditions, and spatiotemporal data. RF was separately trained with the meteorological data (details in Section 2.3) and fine-tuned using a 5-fold cross-validation (CV) technique. We utilized the ResNet50 architecture (He et al., 2016) as the underlying CNN model. The final predictions of the pipeline were obtained by summing the weighted $PM_{2.5}$ values predicted by the RF model and the $PM_{2.5}$ values predicted by the CNN model. This approach forces the CNN model to train for predicting the residual $PM_{2.5}$ hence possible of improving the results. Also, this approach is much easier to train.

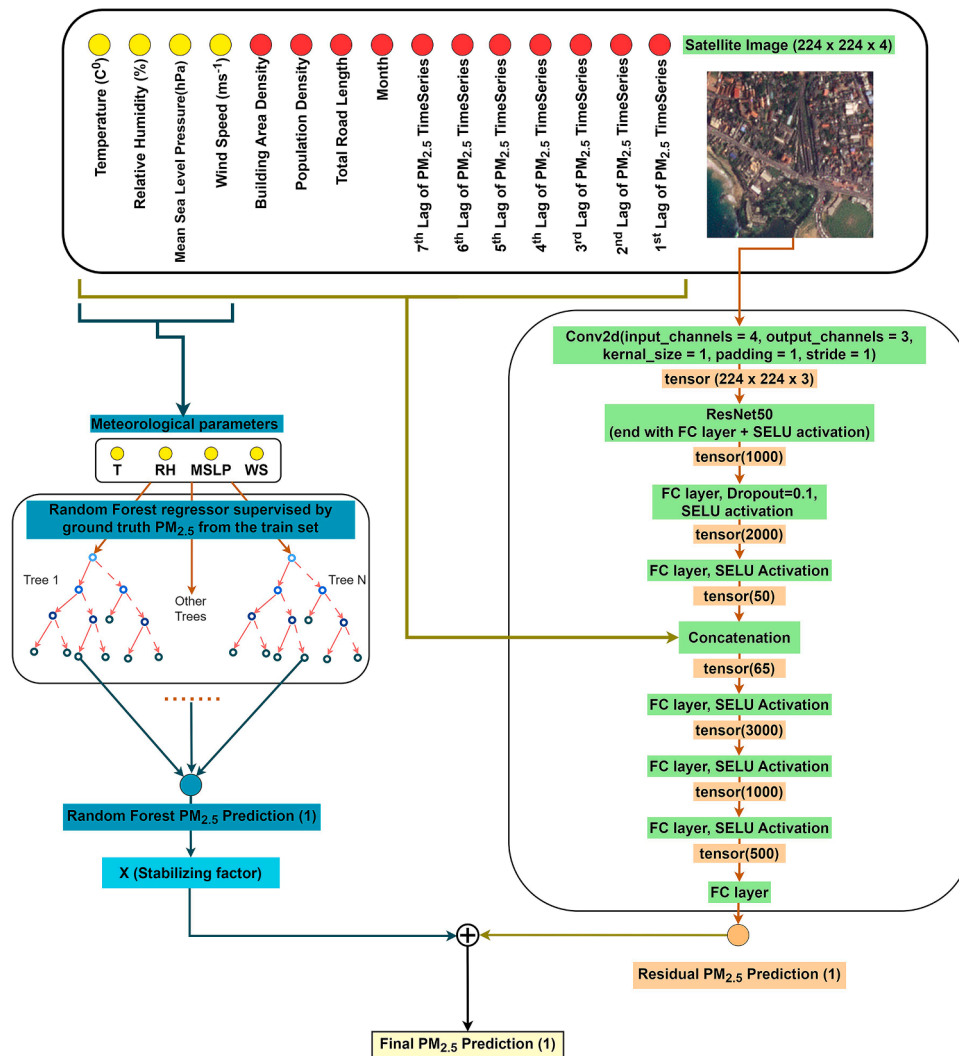


Fig. 7. The complete RF-CNN pipeline including the RF regressor to predict daily average $PM_{2.5}$ using T, RH, MSLP, and WS; and the CNN model trained to predict daily average $PM_{2.5}$ using satellite images, meteorological features, and spatiotemporal features.

2.5.1. Random Forest (RF) model

The primary reason for using Random Forest (RF) for this study is its ability to accurately represent complex and nonlinear connections between input features and the target variable, without relying on specific assumptions about the data distribution (Auret and Aldrich, 2012). During the initial phase of the pipeline, the RF model was trained using ground-truth daily average PM_{2.5} concentrations of all the data points with inputs as meteorological features; T, RH, MSLP, and WS. Each input feature was standardized to have zero mean and standard deviation of one. The RF model was specifically trained using grid search and the 5-fold cross-validation technique. A grid search was conducted to determine the optimal hyperparameters for the RF model. The hyperparameters considered were the number of estimators (N), the maximum depth of a tree (k), and the minimum number of samples required to divide an internal node (m). The grid search was conducted with N in (100, 200, 300, 400, 600), k in (10, 15, 20, 25), and m in (2, 5, 10). The selection of the best hyperparameters was based on the results of 5-fold cross-validation. The number of input features for each tree was set to $\sqrt{4} = 2$. The best choices for N, k, and m were 600, 10, and 2 respectively.

Before training, it is crucial to rescale the RF predictions before they are fed into the CNN component. The scaling factor mentioned by Zheng et al. (2021) is referred to as the 'stabilizing factor'. The selection of the stabilizing factor (SF) should be done with careful consideration to enhance the model more effectively.

2.5.2. CNN model

First, each satellite image of size 224 (height in px) x 224 (width in px) x 4 (number of channels) was fed into the initial CNN layer for the conversion of images to be compatible with the ResNet50 architecture. Then the outputs from the first CNN layer were forwarded through ResNet50 which had the Scaled Exponential Linear Unit (SELU) at the final fully-connected (FC) layer. The 1000 extracted features from the satellite images using ResNet50 were compacted to 50 using another two FC layers with the SELU activation function. Next, this flatten-tensor was concatenated with the meteorological, and spatiotemporal features mentioned in Sections 2.3 and 2.4. Subsequently, this concatenated tensor was passed through three FC layers with SELU activation and finally FC layer of dimension one gives the residual prediction from the CNN side. The summation of weighted RF prediction and CNN residual prediction gives the final prediction result of the RF-CNN pipeline. This RF-CNN model was jointly trained using the Adam optimization algorithm (Kingma and Ba, 2014) with $\beta_1 = 0.9$, $\beta_2 = 0.9$, and $\alpha = 0.001$ to reduce the mean squared error (MSE) between the ultimate predictions of the RF-CNN model and the ground-truth PM_{2.5} concentrations in the train set with the batch size of 32. A multi-step learning rate scheduler was used to reduce the learning rate step-wise. To obtain the best results for each stabilizing factor, a suitable learning rate (see Table 4) was used instead of a common learning rate. In our method, we refrain from utilizing transfer learning and prefer to train the complete deep CNN architecture to specifically suit our purpose. This approach guarantees superior feature extraction from the satellite images compared to transfer learning. Instead of utilizing batch normalization, we choose to use the SELU activation function, which performs internal normalization and does not suffer from the vanishing gradient issue encountered with the Rectified Linear Unit (ReLU). It is important to ensure that the CNN side has sufficient complexity to accurately capture the randomness present in the training set.

2.5.3. Evaluation

The forecasting capabilities of the RF-CNN cascaded model were assessed using a test set consisting of 7 AQM stations, covering different climate zones of Sri Lanka: Galle, Mannar, PointPedro, Kanthale, Ampara, Katubedda, and Kurunegala. First, all the input features undergo the RF-CNN cascaded model, and the resulting predictions are

then evaluated against the ground-truth PM_{2.5} values using the metrics of Pearson correlation coefficients (Pearson r), Spearman correlation coefficient (Spearman r), root mean square error (RMSE), normalized RMSE (NRMSE), mean absolute error (MAE) and mean absolute percentage error (MAPE). The Eqs. (1)-(4) define the terms RMSE, NRMSE, MAE, and MAPE respectively. The Spatial Pearson correlation coefficient (Spatial r) was employed to objectively assess the spatial predictability of the model. The term refers to the 'Spatial r' calculated between the average ground truth PM_{2.5} values for each station and the average PM_{2.5} forecast values for each station (Zheng et al., 2021).

$$RMSE = \sqrt{\frac{1}{N} \left(\|y - \hat{y}\|^2 \right)} \quad (1)$$

$$NRMSE = \frac{RMSE}{\text{Mean PM}_{2.5} \text{ concentration in test set}} \times 100 \quad (2)$$

$$MAE = \frac{1}{N} \|y - \hat{y}\| \quad (3)$$

$$MAPE = \frac{100}{N} \sum_{i=1}^N \frac{\|y - \hat{y}\|}{y} \quad (4)$$

Where y, $\hat{y}$, and N are the ground-truth daily average PM_{2.5} concentrations, PM_{2.5} predictions from the RF-CNN pipeline, and the number of data points in the test set.

2.5.4. Sensitivity to data imputation

Imputed data do not represent the actual values. They will inevitably contain discrepancies when compared to the reality. Therefore, the process of imputation will lead to a deterioration in the quality of the data. Since the data imputation is done on an hourly basis, each imputed data entry in the hourly ground-truth PM_{2.5} concentrations, T, and RH time series can be considered as a single point of data quality degradation. When the data for a specific feature is imputed for an entire day, it can be regarded as 24 data quality degradation points.

The ultimate dataset consists of 9 input features that may require imputation; the ground-truth daily average PM_{2.5} concentrations of the past 7 days, daily average temperature, and daily average relative humidity. For a single data point, a feature value with the highest data quality would have zero degradation points, while a feature value with the lowest data quality would have 24 degradation points. Assuming that 75 % of the hourly data within a day is required to accurately calculate a daily average measurement, a single feature value with degradation points of up to 6 can be considered as good quality for both input and testing quality.

When we calculated the overall input data quality of a data point, it was essential to consider the data quality of all the input feature values that could be subjected to imputation. Simply calculating the mean of the degradation points of the input feature values that may undergo imputation can lead to substantial challenges in recognizing the overall quality of the input data. The Eq. (5) demonstrates our approach to resolving those concerns through the introduction of the Input Data Quality Score (IDQS). It is important to note that Eq. (5) calculates the IDQS data point by point, rather than feature by feature.

$$IDQS = \frac{\sum_i^k \min(0, D_a - x_i)}{D_m - D_a} + k \quad (5)$$

D_a , D_m , k , and x_i represent the maximum allowable degradation points for an input feature value to classify it as a feature value with good data quality, the maximum degradation points that an input feature can have for a single feature value, the number of input features subjected to imputation, and degradation points of i^{th} input feature value in the selected single data point respectively. Therefore, the IDQS has a numerical range from 0 to k , with k representing the highest possible IDQS value. A higher IDQS corresponds to a higher level of

input data quality of a data point. In this scenario; D_w , D_m , and k will be 6, 24, and 9 respectively. This approach can also be utilized for computing the Testing Data Quality Score (TDQS). The only difference is that the value of k will be 1 for the TDQS.

3. Results

3.1. Prediction performances of the RF

As stated in Section 2.5.1, the best hyperparameters were chosen using grid search and within the grid search, the mean 5-fold cross-validation result was utilized as the results of each search. Table 3 shows the 5-fold CV results for the optimal hyperparameters mentioned in 2.5.1. Fig. 8 shows the $PM_{2.5}$ prediction performance of the RF part using the test set. According to Table 4, RF yields high prediction errors in the test set with an NRMSE, RMSE, MAPE, and MAE of 55.922 %, 8.151 $\mu g/m^3$, 52.084 %, and 5.791 $\mu g/m^3$, respectively. Pearson r and Spearman r are 0.519 and 0.471 respectively. Spatial r resulted in a low value of 0.412 with a p-value of 0.358. During the process of learning the decision rules for each estimator in RF, MSLP held a crucial role in reducing the entropy of the data at any decision node. The other significant factors that contributed to the reduction of impurities, listed in descending order, were T, RH, and WS. Since the prediction capability of the RF part was low, the stabilizing factor should be a low value to reduce the error propagating to the cascaded model and to have enough room to capture the spatial variability. Proposed modifications and alterations to the cascaded model will improve its predictive capabilities and yield more precise outcomes.

3.2. Prediction performances of the RF-CNN cascaded model

According to Table 3 and Fig. 9, it can be realized that the RF predictions are not sufficiently accurate to be given greater importance in the cascaded model. Therefore, it was thought that an improved outcome would be achieved by setting the stabilizing factor to zero, thereby eliminating the RF component. This might be crucial because the presence of an RF component would introduce a significant error to the CNN component, which is unable to effectively compensate for such high levels of error due to the change in the dynamics between input features and the target variable. Hence to determine the optimal stabilizing factor for the RF-CNN cascaded model, three different factors were used during training: 0.0, 0.1, and 0.3. The selection of the stabilizing factor greatly impacts the final performance of the model. Table 4 presents the testing outcomes for each stabilizing factor.

The data presented in Table 4 clearly demonstrates that the CNN model (RF-CNN cascaded model with a stabilizing factor of 0.0) has lower prediction performance compared to the RF-CNN cascaded model with a stabilizing factor of 0.1. Therefore, with a low weight, the RF component has contributed to enhancing the accuracy of predictions in the cascaded model. We could say that the RF part manipulates the $PM_{2.5}$ dynamics in a specific manner allowing the cascaded model to effectively capture the dependencies with its input features. Therefore, the role of the RF component in our model is distinct from that in Zheng et al. (2021) (see Section 1).

The RF-CNN cascaded model exhibits a significant improvement in

Table 3

RMSE, NRMSE, MAE, MAPE, and Pearson r of the 5-fold CV of the RF model for the optimal hyperparameters.

Fold	RMSE ($\mu g/m^3$)	NRMSE	MAE ($\mu g/m^3$)	MAPE	Pearson r
1	8.729	56.460 %	6.525	53.201 %	0.521
2	7.079	50.643 %	5.191	55.077 %	0.597
3	8.119	54.502 %	6.264	63.984 %	0.554
4	8.544	56.748 %	6.274	64.289 %	0.607
5	8.060	53.159 %	6.090	55.481 %	0.569

prediction performance compared to the RF model. Table 4 indicates that the RF-CNN cascaded model, which utilizes a stabilizing factor of 0.1, gives the most optimal outcomes. It was trained with 15 epochs and an initial learning rate of $1e^{-5}$. At the 3rd and 10th epochs, the learning rate was reduced by a factor of $1e^{-3}$. This approach yields comparatively low prediction errors and good correlation coefficients in the test set with RMSE, NRMSE, MAE, MAPE, Spearman r , and Pearson r of 4.726 $\mu g/m^3$, 32.422 %, 3.318 $\mu g/m^3$, 25.691 %, 0.871 and 0.873, respectively. Fig. 8 displays the scatterplot comparison of the test results for both the optimal RF-CNN model and the RF model. The optimal RF-CNN cascaded model has significantly improved the spatial predictions, demonstrating the precise incorporation of additional information from the CNN component of the model. It exhibits a spatial r of 0.978 with a p-value of 0.00013. Fig. 8 also shows the scatterplots corresponding to the spatial r calculation of the test results of both the RF and RF-CNN models.

The prediction results of the optimal cascaded model for each test station are presented in Table 5. Predictions for all the test stations had NRMSE of <42 % and MAPE of <31 %. Kurunegala has the lowest NRMSE of 25.669 %, while Ampara has the lowest MAPE of 22.079 %. The correlation metrics: Spearman r and Pearson r exhibit values significantly exceeding 0.74 across the majority of the test sites. The Katubedda site exhibits a correlation slightly below 0.75, along with relatively higher values of RMSE and MAE compared to the results obtained from other test stations. Fig. 9 illustrates the scatterplot comparison of the test results for each test station.

3.3. Effect of the imputation on prediction evaluation

Within the training set, 92 % of the data points have an IDQS higher than 6.75, 5 % have an IDQS of 0, 92 % of the data points have a TDQS higher than 0.75, while 6.5 % of the data points have a TDQS of 0. In the test set, almost 95 % of the data points have an IDQS higher than 6.75, and 2 % have an IDQS of 0. Also, 96 % of the data points in the test set have a TDQS higher than 0.75 while 3.5 % of the data points have a TDQS of 0. Therefore, it is certain that a small portion of the data points in both the train and test sets are completely derived from the imputation process, while the majority of the data points exhibit superior data quality. Fig. 10 provides additional details regarding the distributions of IDQS and TDQS in both the train and test sets.

With an IDQS of 9, the cascaded model performs well since all the input features have the best data quality and it could test with the data points with TDQS of 1. Allowing IDQS values higher than 6.75 still results in satisfactory performance. When the data points have IDQS higher than 6.75 but low TDQS (<0.75), the performance metrics indicate higher errors, highlighting the significance of high TDQS when evaluating the performance. When the IDQS is lower than 6.75, the cascaded model performs moderately indicating it might work well with low $PM_{2.5}$ concentrations but not with moderate levels of $PM_{2.5}$ concentrations. When both the IDQS and TDQS are low, the error metrics indicate better performance. However, the evaluation is uncertain because of the low TDQS of the data points. Fig. 11 illustrates the performance of the cascaded model when the IDQS and TDQS changes. Based on Fig. 11, the imputation process is effective in filling the missing data in the input feature space. If the evaluation metrics show low errors with higher IDQS and low TDQS, we may suggest using the employed imputation method to fill in the gaps in the target $PM_{2.5}$ concentrations which could be useful to assess the performance of the model accurately enough.

3.4. Sensitivity to the satellite images

Using different satellite images captured at various times within the same day can result in slight variations in the predictions for that day. This is completely due to the difference in the features extracted from the CNN part to predict the final results. The predictions exhibit an

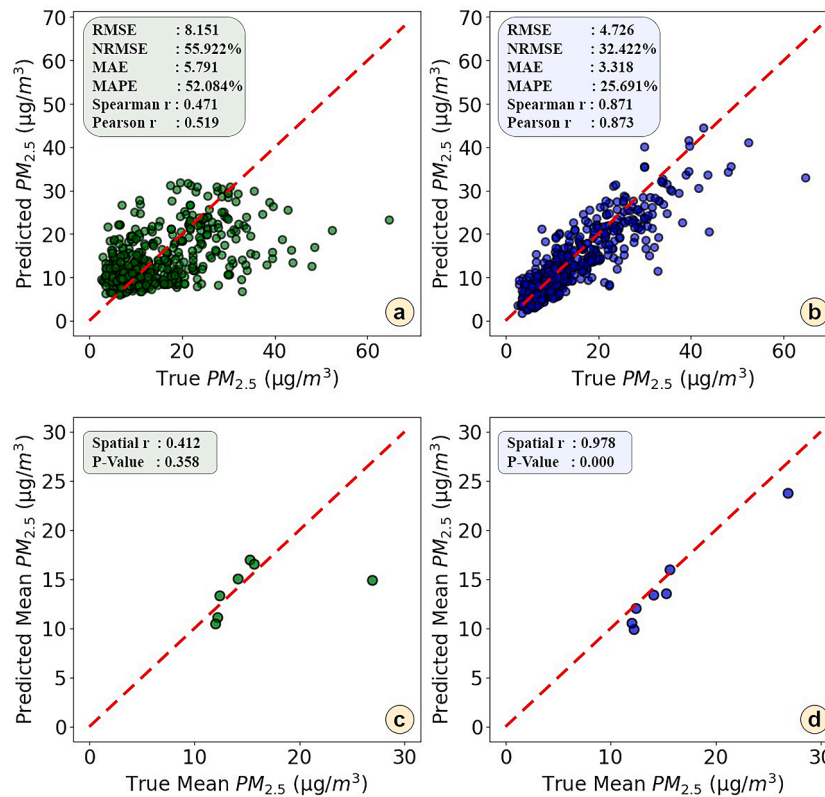


Fig. 8. Scatterplots of a) the predicted $PM_{2.5}$ concentrations in the test set from RF against the ground-truth $PM_{2.5}$ concentrations in the test set; b) the predicted $PM_{2.5}$ concentrations in the test set from RF-CNN cascaded model against the ground-truth $PM_{2.5}$ concentrations in the test set; c) station wise predicted mean $PM_{2.5}$ concentrations in the test set from RF against the station wise ground-truth mean $PM_{2.5}$ concentrations in the test set; d) station wise predicted mean $PM_{2.5}$ concentrations in the test set from RF-CNN cascaded model against the station wise ground-truth mean $PM_{2.5}$ concentrations in the test set. Note that the red dashed lines are the 1:1 line.

Table 4

RMSE, NRMSE, MAE, MAPE, Spearman r, and Pearson r of the testing set results for the RF model, RF-CNN cascaded model with stabilizing factor of 0.0, RF-CNN cascaded model with stabilizing factor of 0.1, and RF-CNN cascaded model with stabilizing factor of 0.3. The learning rate for each cascaded model is also included.

Model	Learning Rate	RMSE ($\mu g/m^3$)	NRMSE	MAE ($\mu g/m^3$)	MAPE	Spearman r	Pearson r
RF	–	8.151	55.922 %	5.791	52.084 %	0.471	0.519
RF-CNN with SF=0.0	8e-5	4.801	32.941 %	3.345	26.041 %	0.874	0.860
RF-CNN with SF=0.1	1e-5	4.726	32.422 %	3.318	25.691 %	0.871	0.873
RF-CNN with SF=0.3	2e-5	4.828	33.121 %	3.450	29.934 %	0.859	0.862

average deviation of 7.988 % compared to its ground-truth $PM_{2.5}$ concentrations. Fig. 12 illustrates four examples that demonstrate the impact of satellite images on the final predictions. As shown in Fig. 12, the images exhibit minimal variation, primarily in terms of their warmth, subtle changes in color, slightly blurred regions, and areas with shadows. Aside from the blurriness, other characteristics have the potential to significantly influence the outcome of the predictions.

4. Discussion

This study represents the first comprehensive investigation into the spatial variations of $PM_{2.5}$ concentrations throughout the country. It aims to predict daily average $PM_{2.5}$ concentrations across Sri Lanka using a cascaded RF-CNN pipeline that leverages meteorological data, spatiotemporal data, and high-resolution multiband satellite images. After testing the proposed approach for 7 AQM stations covering different climate zones of Sri Lanka, we have determined that the RF-CNN cascade model is capable of forecasting $PM_{2.5}$ concentrations accurately enough throughout the country. This study employs K-Nearest Neighbors data imputation on both the input feature space and

the target $PM_{2.5}$ concentrations. Using completely ground-truth data without imputation, the cascaded model was able to predict the $PM_{2.5}$ concentrations with low prediction errors in the test set. The cascaded model exhibited a negligible decline in prediction performance with the allowed levels of imputation on the ground-truth data. Furthermore, the cascaded model produces satisfactory results even when the input features have lower data quality than what is permitted. Hence it may be advisable to employ the KNN imputation method to fill in the missing values in the input feature space. However, a comprehensive investigation is required to verify the scalability of this choice. Using ground-truth data with imputation levels below the allowed levels as target values to evaluate performance is not advisable.

The RF predictions exhibit high errors due to the inadequate precision of the relative humidity and temperature measurements from the BlueSky device. However, the RF-CNN pipeline yielded satisfactory outcomes by employing the RF component as an error model. With a stabilizing factor of 0.1, the RF component facilitates the manipulation of $PM_{2.5}$ dynamics to enhance the CNN component's ability to accurately predict the residual. Hence the RF part could be described as a successful error model.

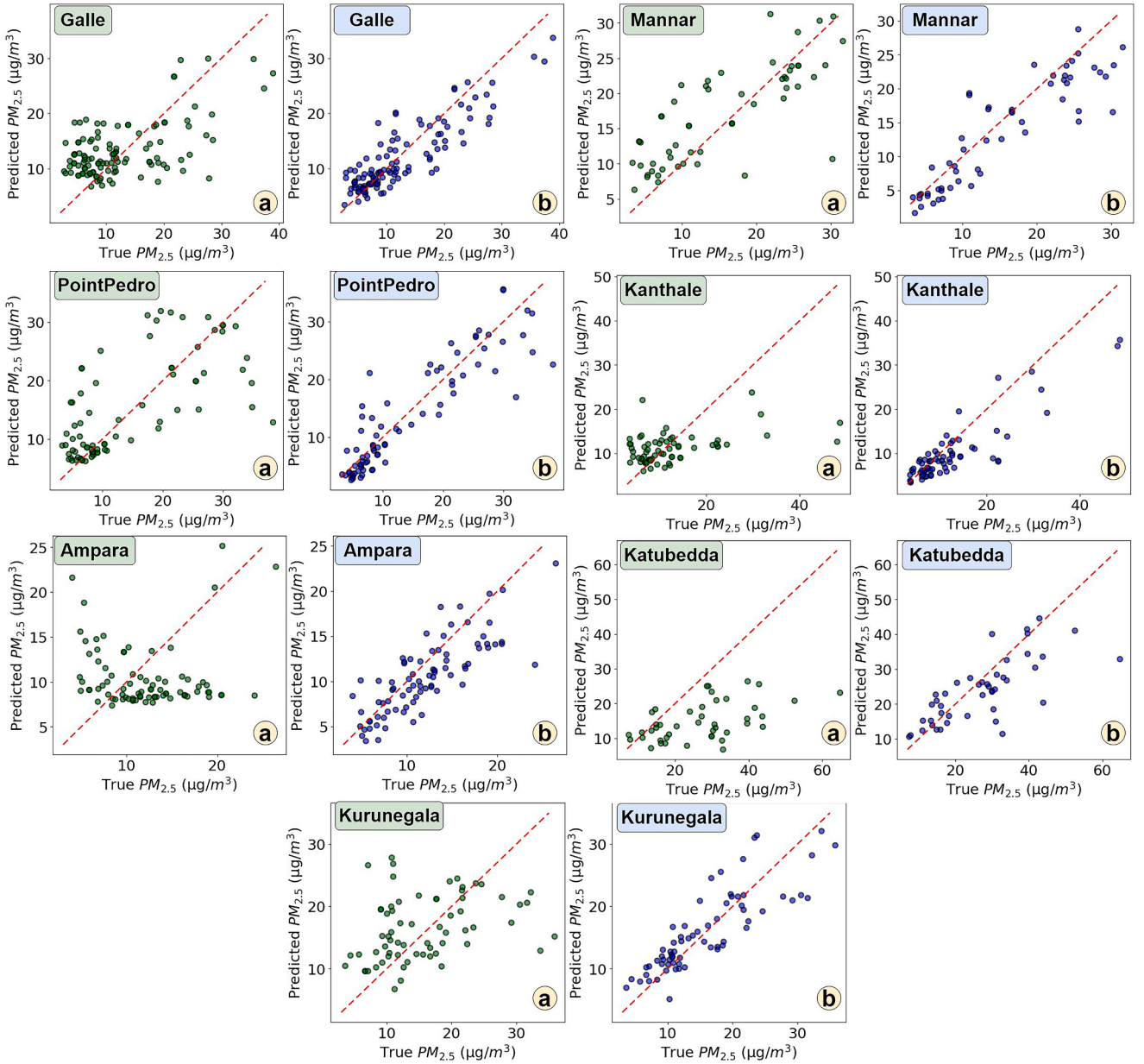


Fig. 9. Scatterplots of a) the ground-truth $PM_{2.5}$ concentrations against the predicted $PM_{2.5}$ concentrations from RF for each test site and b) the ground-truth $PM_{2.5}$ concentrations against the predicted $PM_{2.5}$ concentrations from RF-CNN cascaded model for each test site. Note that corresponding test sites are labeled accordingly on the scatterplot and the red dashed lines are the 1:1 line.

According to the findings of Zheng et al. (2020) which used a CNN-RF cascaded model, the Beijing test dataset yields MAE, NMAE, RMSE, and NRMSE of $10.5 \mu\text{g}/\text{m}^3$, 24.6 %, $19.1 \mu\text{g}/\text{m}^3$, and 44.7 %, respectively. Also, Zheng et al. (2020), the Shanghai test dataset showcased MAE, NMAE, RMSE, and NRMSE of $7.7 \mu\text{g}/\text{m}^3$, 18.6 %, $10.8 \mu\text{g}/\text{m}^3$, and 26.0 %, respectively. Based on the findings of Zheng et al. (2021), they have successfully predicted $PM_{2.5}$ concentrations in Delhi with an RMSE, MAE, NRMSE, and NMAE of $32.69 \mu\text{g}/\text{m}^3$, $20.07 \mu\text{g}/\text{m}^3$, 31.1 %, and 19.1 %, respectively, using RF-CNN approach. Also, they have predicted $PM_{2.5}$ concentrations in Beijing with the RMSE, MAE, NRMSE, and NMAE of $13.00 \mu\text{g}/\text{m}^3$, $7.26 \mu\text{g}/\text{m}^3$, 30.4 %, and 17.0 %. It is important to note that the Delhi, Beijing, and Shanghai datasets have an overall average of $PM_{2.5}$ concentration well above $50 \mu\text{g}/\text{m}^3$. Our dataset has an overall average of $PM_{2.5}$ concentration around $15 \mu\text{g}/\text{m}^3$ and exhibits a similar prediction performance with RMSE, NRMSE, MAE, and NMAE of $4.726 \mu\text{g}/\text{m}^3$, 32.422 %, $3.318 \mu\text{g}/\text{m}^3$, and 22.760 %, respectively. Our method, which incorporates a low stabilizing

factor, and additional spatiotemporal input features, achieved higher spatial predictability compared to the approach described in Zheng et al. (2021), with a spatial r of 0.978. This study showcases the ability of the RF-CNN pipeline, even in situations where $PM_{2.5}$ concentration levels are relatively lower. While our approach differs slightly from Zheng et al. (2021), having low stabilizing factor and the inclusion of spatiotemporal features substantially enhanced the model performance.

Traditional government stations use high-precision instruments, yielding superior accuracy exceeds US-EPA Class III $PM_{2.5}$ FEM standards for additive and multiplicative bias but at a high cost for dense spatial deployment around \$4500 per station. In contrast, the proposed solution achieves moderate yet sufficient accuracy while significantly reducing the establishing and maintaining cost. One low-cost sensor costs around \$350. Instead of satellite images, it is possible to use images acquired by drones. Since the RF-CNN is a moderately complex model, it consumes low power and to further decrease in energy consumption, depthwise-separable ResNet architecture can be employed as the CNN

Table 5

RMSE, NRMSE, MAE, MAPE, Spearman r, and Pearson r of the test results for each test station using the RF model and the RF-CNN cascaded model with stabilizing factor of 0.1.

Test Station	Model	RMSE ($\mu\text{g}/\text{m}^3$)	NRMSE	MAE ($\mu\text{g}/\text{m}^3$)	MAPE	Spearman r	Pearson r
Galle	RF	6.536	52.651 %	5.246	63.747 %	0.355	0.573
	RF-CNN	3.879	31.246 %	3.000	30.974 %	0.817	0.876
Mannar	RF	5.721	37.429 %	4.318	46.959 %	0.763	0.773
	RF-CNN	4.220	27.609 %	3.108	22.901 %	0.888	0.894
Point-Pedro	RF	7.332	52.006 %	5.015	49.438 %	0.648	0.686
	RF-CNN	4.347	30.831 %	3.093	27.104 %	0.896	0.899
Kanthale	RF	8.239	67.547 %	5.321	53.153 %	0.348	0.458
	RF-CNN	5.070	41.568 %	3.415	25.830 %	0.778	0.882
Ampara	RF	6.142	51.228 %	4.945	50.096 %	-0.175	0.011
	RF-CNN	3.219	26.849 %	2.525	22.079 %	0.816	0.804
Katubedda	RF	16.013	59.563 %	12.602	41.520 %	0.461	0.519
	RF-CNN	8.900	33.100 %	6.270	23.611 %	0.746	0.740
Kurunegala	RF	7.608	48.636 %	5.870	48.039 %	0.396	0.325
	RF-CNN	4.015	25.669 %	3.212	23.055 %	0.879	0.845

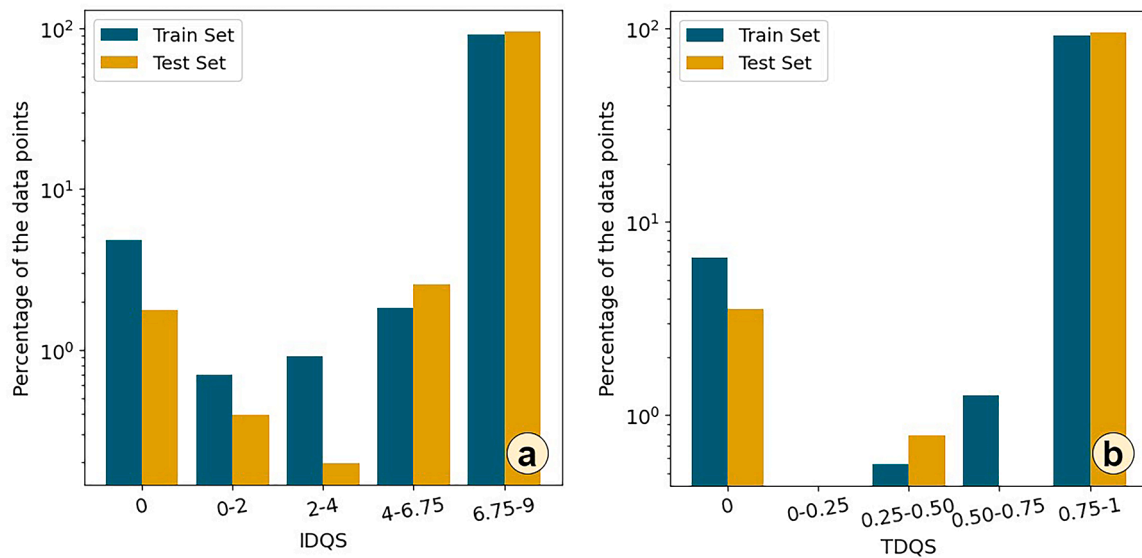


Fig. 10. Histogram plots of a) IDQS distribution in both train and test sets; b) TDQS distribution in both train and test sets. The y-axis of each histogram plot is represented on a logarithmic scale.

model to process the satellite images. The trade-off lies in tolerating marginally increased uncertainty in return for enhanced spatial coverage and cost-effectiveness, making it especially suitable for resource-constrained areas.

The applicability of the proposed model to areas with lack of standard monitoring has several implementation steps. First using carefully deployed low-cost sensor network, a ground-level $\text{PM}_{2.5}$ dataset should be created with enough resolution. Utilizing the gathered data, the clustering of sensor nodes and the training of the RF-CNN pipeline can be carried out. During the clustering process, each low-cost sensor station will be allocated to a cluster according to its meteorological patterns and $\text{PM}_{2.5}$ concentration variation. Consequently, each cluster will possess a distinct, non-overlapping geographical area. Then suggested RF-CNN model can forecast $\text{PM}_{2.5}$ concentrations in locations lacking monitoring. The temporal input features required to produce a model output can be derived using spatial interpolation techniques, such as Nearest Neighbor, Inverse Distance Weighting, or the Kriging algorithm, inside the relevant cluster. The sensor network must be adequately maintained for further finetuning the system. The model's sustained resilience to climate change relies on its ability to adapt to changing environmental conditions. Meteorological features (temperature, humidity, wind speed, etc.) are essential for forecasts; nevertheless,

alterations in weather pattern dynamics may need retraining the model with an updated dataset. Moreover, climate change could increase transboundary pollution or local emission sources (e.g., wildfires), requiring ongoing model retraining and transfer learning to adjust to evolving pollution dynamics. Therefore, it is essential to maintain a dataset for the purpose of enhancing the model in the future.

The satellite images taken on the same day may vary slightly depending on the time it captured. Therefore, having a significant number of days with multiple satellite images may be beneficial in enhancing the model's generalizability and robustness. The CNN component exhibits a slight sensitivity to satellite images containing undesirable features such as clouds and shadowy areas due to the fact that the majority of the dataset consists of clear satellite images. It is important to acknowledge that even in two distinct satellite images taken 10 min apart, there may be observable inconsistencies. Therefore, it is crucial in future studies to gather satellite images, including any potential variations that may be present, in order to enhance the resilience of the model. The RF model only utilizes the temperature, relative humidity, wind speed, and mean sea level pressure as the input features. It might be worthwhile to explore how other highly influential meteorological factors, such as precipitation, affect the accuracy of predictions for $\text{PM}_{2.5}$ concentrations in both the RF model and the cascaded model.

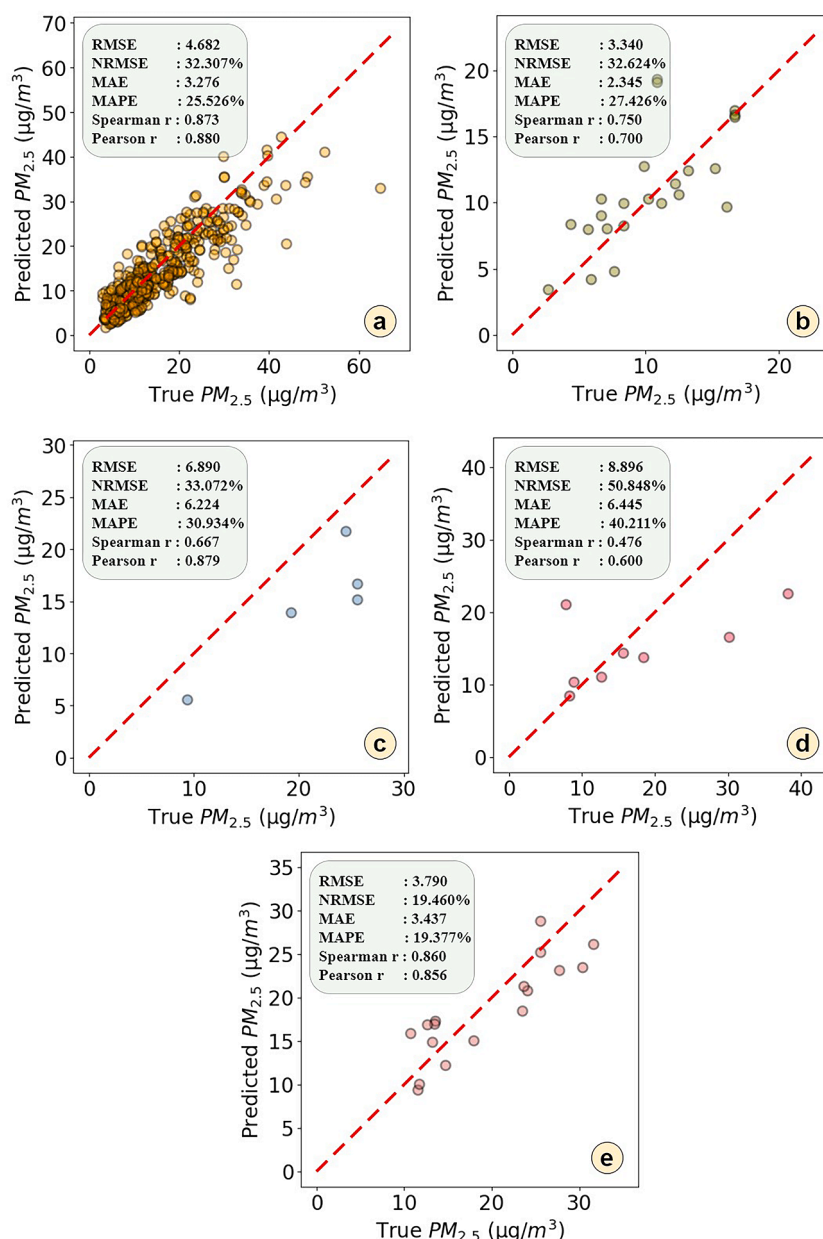


Fig. 11. Scatterplots of a) the predicted $PM_{2.5}$ concentrations in the test set with IDQS of 9 from the RF-CNN cascaded model against the ground-truth $PM_{2.5}$ concentrations with TDQS of 1 in the test set; b) the predicted $PM_{2.5}$ concentrations in the test set with $6.75 < IDQS < 9$ from the RF-CNN cascaded model against the ground-truth $PM_{2.5}$ concentrations with $0.75 < TDQS < 1$ in the test set; c) the predicted $PM_{2.5}$ concentrations in the test set with IDQS higher than 6.75 from the RF-CNN cascaded model against the ground-truth $PM_{2.5}$ concentrations with TDQS below 0.75 in the test set; d) the predicted $PM_{2.5}$ concentrations in the test set with IDQS below 6.75 from the RF-CNN cascaded model against the ground-truth $PM_{2.5}$ concentrations with TDQS higher than 0.75 in the test set; and e) the predicted $PM_{2.5}$ concentrations in the test set with IDQS below 6.75 from the RF-CNN cascaded model against the ground-truth $PM_{2.5}$ concentrations with TDQS below 0.75 in the test set.

The existing methodology has several limitations, mainly insufficient sensor coverage and partially addressed seasonality of the dataset. This limits the model's applicability in regions without adequate ground-truth data. Furthermore, the manual preprocessing of satellite imagery involves subjectivity and lacks scalability for extensive datasets or real-time applications. Also, the model is very susceptible to image quality, especially with clouds and shadows.

This study marks a crucial advancement in forecasting $PM_{2.5}$ concentrations throughout Sri Lanka; however, it is crucial to resolve its limitations to enhance its application in different locations and reach higher robustness. Future efforts must emphasize scalability, automation, and the integration of multiple data sources to establish a thorough and adaptable framework for air quality monitoring. To cover the long-

term trends, a multi-year dataset should be maintained. To enhance coverage in sensor-deficient areas, synthetically generated data can replicate $PM_{2.5}$ values by employing data assimilation techniques and interpolation from adjacent locations. In areas with inadequate sensor coverage, predictions can be enhanced by integrating high-resolution satellite data, including multispectral and hyperspectral bands. The RF-CNN pipeline can be improved using incomplete data from new locations and incorporate existing high-quality datasets via transfer learning, enhancing performance across diverse areas. Automating the preprocessing of satellite images can be achieved using deep learning-based segmentation algorithms to efficiently identify and mask clouds and shadows. Moreover, the implementation of dynamic thresholds for dark pixel and cloud coverage filters facilitates adaptive preprocessing

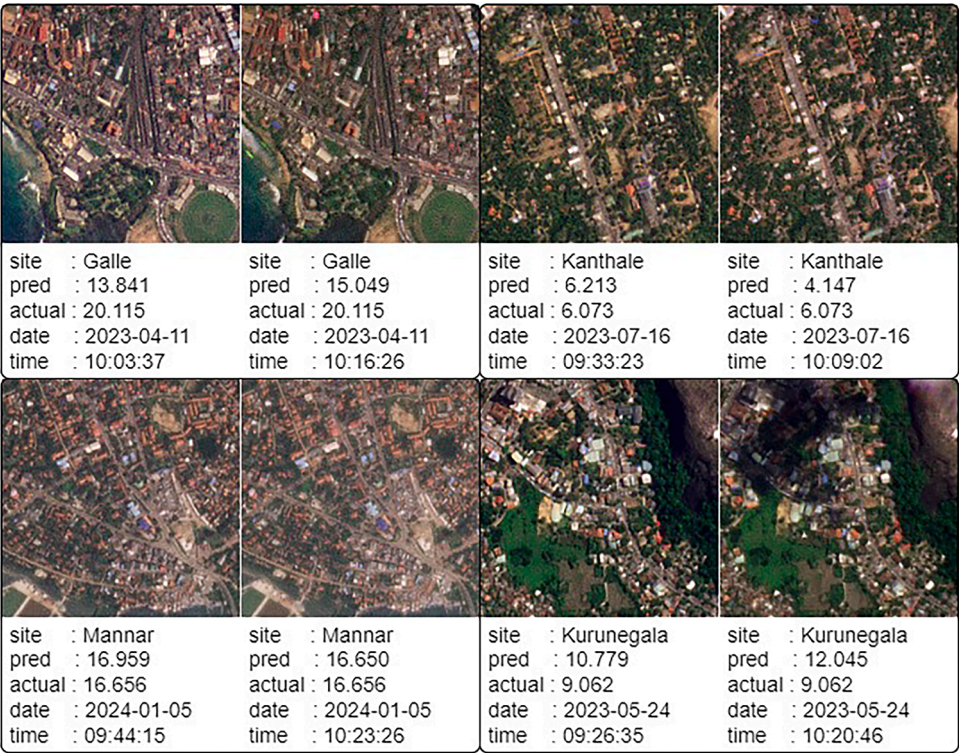


Fig. 12. Two satellite images captured at different times on a single day for the Galle, Kanthale, Mannar, and Kurunegala AQM sites. Corresponding AQM site (site), prediction from the RF-CNN cascaded model (pred), daily average ground-truth PM_{2.5} concentration (actual), date of the satellite image captured (date), and the timestamp of when the satellite image was taken (time) are clearly labeled under each satellite image.

adapted to geographical and temporal fluctuations, hence providing enhanced consistency and accuracy of data quality at scale. Enhancing the model’s generalizability requires the expansion of spatiotemporal features such as land use data, vegetation indices, and traffic patterns to enhance regional adaptability. Also, data augmentation can enhance the generalizability by incorporating synthetic variations in satellite imagery, replicating various lighting conditions, cloud scenarios, and seasonal changes in the training dataset to improve predictive performance. Furthermore, investigating the transferability of the RF-CNN pipeline to other pollutants, such as NO₂ or O₃, can further demonstrate its versatility and wider applicability.

5. Conclusion

The RF-CNN pipeline can serve as a robust and comprehensive method for precisely forecasting the spatial and temporal fluctuations in global PM_{2.5} concentrations. Its utility extends beyond simple forecasting, establishing a foundation for improved air quality management worldwide. An essential step in utilizing this model for worldwide application is the identification of regions exhibiting similar PM_{2.5} and meteorological trends, facilitating the categorization of areas into clusters with common dynamics. This area segmentation facilitates the optimization of the RF-CNN configuration to address specific PM_{2.5} dynamics and unique environmental conditions, thus enhancing forecast accuracy and model adaptability. Future extensions of this research may concentrate on integrating supplementary data sources, including real-time emissions data, land use information, and socio-economic factors to enhance the model’s inputs and further optimize its outputs. Moreover, collaborative efforts to develop standardized, high-quality datasets across regions could substantially improve the global applicability of the RF-CNN methodology. With its combination of state-of-the-art machine learning methods, domain-specific knowledge, and global datasets, the RF-CNN pipeline is intended to enhance air quality forecasting and guide policymakers in mitigating air pollution impacts

worldwide.

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CRedit authorship contribution statement

Gimhan Attanayake: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Mahesh Senarathna:** Writing – original draft, Visualization, Validation, Project administration, Investigation, Formal analysis, Data curation. **Mike Bergin:** Writing – review & editing, Supervision, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **David Carlson:** Writing – review & editing, Supervision, Resources, Methodology, Investigation. **Prakash V Bhawe:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization. **Gayan Bowatte:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Conceptualization. **Nalin Harischandra:** Writing – review & editing, Supervision, Methodology, Investigation, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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